

WASP-163b

R-2228

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-163_250323 · created 2026-08-15T00:50:02+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460758.006 +/- 0.055 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.14 +/- 0.042
Transit depth (Rp/Rs)^2	1.97 +/- 1.18 [%]
Orbital Inclination (inc)	83.36 +/- 2.76
Airmass coefficient 1 (a1)	1.59 +/- 0.019
Airmass coefficient 2 (a2)	-0.326 +/- 0.077
Scatter in the residuals of the lightcurve fit is	1.22 %
Best Comparison Star	#2 - [494, 218]
Optimal Aperture	3.12
Optimal Annulus	11.71
Transit Duration (day)	0.038 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.14 ± 0.042 vs 0.119 (deviation 0.44σ)
Duration fit vs published	0.038 d vs 0.0942 d (deviation 1.83σ)
Mid-time O–C	76.2 min
Depth SNR	2.9
Residual scatter	1.22 %

Light curve

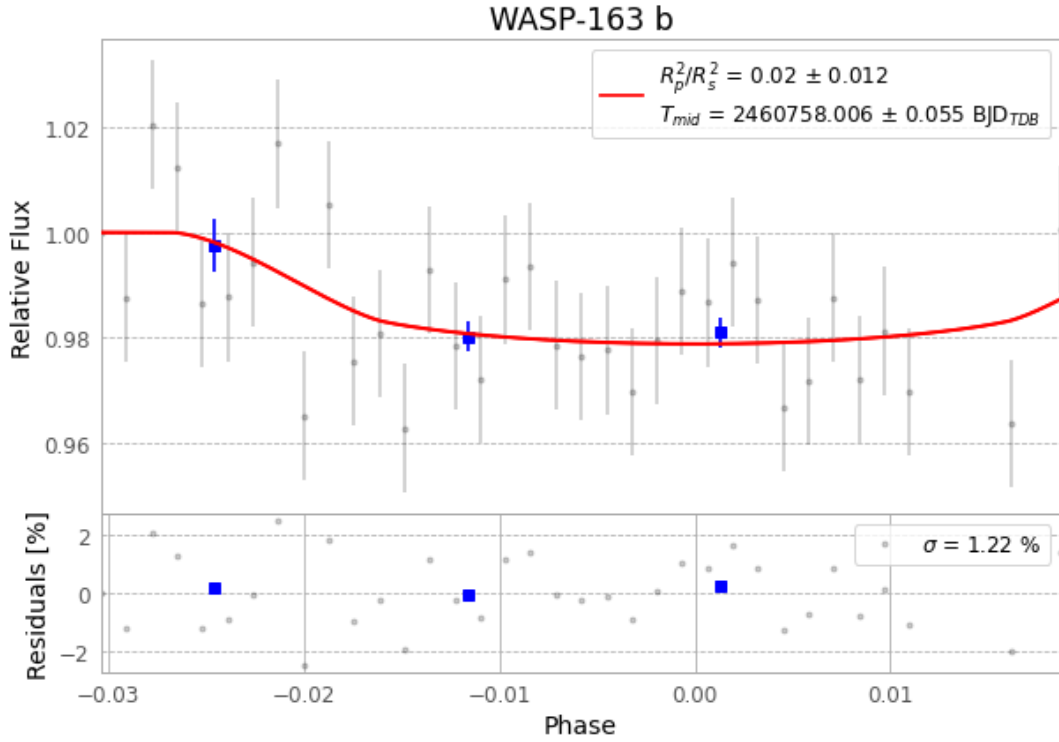


Figure 1 — FinalLightCurve_WASP-163 b_2025-03-23.png (SHA-256 754b0a898a2286a2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [338, 226], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 21be34b30222ab55...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

39 FITS frames (26 MB), WASP-163250323105415.FITS → WASP-163250323130617.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-163-250323.log (51273638ea35...), FinalLightCurve_WASP-163 b_2025-03-23.png (754b0a898a22...), FinalLightCurve_WASP-163 b_2025-03-23.csv (37f8cab2e322...)

Compute resources

Wall time 82.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-15 00:50Z.